

JEE ADVANCED WEEKEND TEST SERIES



NARAYANA
IIT ACADEMY
INDIA



Sec: SR.IIT_*CO-SC(MODEL-A&B)
Time: 3 Hrs

Date: 23-06-24
Max. Marks: 198

23-06-24_SR.IIT_STAR CO-SC(MODEL-A&B)_JEE ADV_WAT-51_SYLLABUS

MATHEMATICS: PRESENT WEEK(80%): Permutations & Combinations: Fundamental

Principles of counting, Linear permutations, Circular permutations

PREVIOUS WEEK(20%): Binomial theorem : Binomial theorem for a positive integral index, Constant term, Numerically greatest term, Properties of the Binomial coefficients, Multinomial Theorem

PHYSICS:

PRESENT WEEK(80%): OPTICS-III : Refraction at curved surface, Lenses and combination of lenses

MAINS EXPERIMENTS: The focal length of Convex lens, using the parallax method.

ADVANCED EXPERIMENTS: focal length of a concave mirror and a convex lens using u-v method

PREVIOUS WEEK(20%): OPTICS-II : Refraction at plane surface, TIR and Prism, dispersion and deviation in thin prism

MAINS EXPERIMENTS: The plot of the angle of deviation vs angle of incidence for a triangular prism.

Refractive index of a glass slab using a travelling microscope.

CHEMISTRY:

PRESENT WEEK(80%): Colour, Magnetic properties, Catalytic properties, alloy and interstitial compounds formation, oxides and oxoanions of metals, Preparation & Properties of Potassium dichromate, F-BLOCK

PREVIOUS WEEK(20%): d-Block elements : General Characteristics, electronic configuration, atomic & ionic size, density, ionisation enthalpy, melting points, enthalpy of atomization, Oxidation states, Chemical reactivity and E0 values, Nature of oxides

PHYSICS

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 1 – 6)	Questions with Single Correct Options	+3	-1	6	18
Sec – II (Q.N : 7 – 12)	One of More Correct Options Type (partial marking scheme) (+1)	+4	-2	6	24
Sec – III (Q.N : 13 – 18)	Questions with Numerical Value Type (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)	+4	0	6	24
Total				18	66

CHEMISTRY

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 19 – 24)	Questions with Single Correct Options	+3	-1	6	18
Sec – II (Q.N : 25 – 30)	One of More Correct Options Type (partial marking scheme) (+1)	+4	-2	6	24
Sec – III (Q.N : 31 – 36)	Questions with Numerical Value Type (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)	+4	0	6	24
Total				18	66

MATHEMATICS

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 37 – 42)	Questions with Single Correct Options	+3	-1	6	18
Sec – II (Q.N : 43 – 48)	One of More Correct Options Type (partial marking scheme) (+1)	+4	-2	6	24
Sec – III (Q.N : 49 – 54)	Questions with Numerical Value Type (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)	+4	0	6	24
Total				18	66

THE NARAYANA GROUP

MATHEMATICS**MAX.MARKS: 66****SECTION – 1 (Maximum Marks: 18)**

This section contains SIX (06) questions.

Each question has FOUR options for correct answer(s). **ONLY ONE** of these four option is the correct answer. For each question, choose the correct option corresponding to the correct answer.

Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If only the correct option is chosen.

Zero Marks: 0 If none of the option is chosen.(i.e the question is un answered)

Negative Marks: -1 In all other cases.

1. The number of ways of forming an arrangement of 5 letters from the letters of the word “IITJEE” is
A) 60 B) 96 C) 120 D) 180
2. The number of times the digit 3 will be written when listing the integers from 1 to 1000, is
A) 269 B) 300 C) 271 D) 302
3. The coefficient of x^{50} in the expansion of $(1+x)^{1000} + 2x(1+x)^{999} + 3x^2(1+x)^{998} + \dots + 1001x^{1000}$ is
A) $^{1000}C_{50}$ B) $^{1001}C_{50}$ C) $^{1002}C_{50}$ D) 2^{1001}
4. A flight of stairs has 10 steps. A person can go up the steps one at a time, two at a time or any combination of 1s and 2s. The total number of ways in which the person can go up the stairs is
A) 75 B) 79 C) 85 D) 89
5. Five boys and four girls are to be seated in a row so that two particular girls will never sit adjacent to a particular boy and all girls are separated. If the number of ways in which they can be seated is N, then value of $\frac{N}{480}$ is-
A) 32 B) 36 C) 40 D) 48
6. The number of three digit numbers with three distinct digits such that one of the digits is the arithmetic mean of the other two is
A) 120 B) 180 C) 112 D) 104

SECTION - 2 (Maximum Marks : 24)

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For each question, choose the correct option(s) to answer the question.

Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks: +3 If all the four options are correct but ONLY three options are chosen.

Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options.

Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks: -2 In all other cases.

7. Let $f(x)$ be a real valued function satisfying $f^3(x) - 6(f(x))^2 + 11f(x) - 6 = 0$ defined on $[0, 8]$. If the function $y = f(x)$ satisfies the condition that $f(x)$ is discontinuous at every integral value of x and continuous at every other x , then the number of such functions is $2^a \times 3^b \times 7^c$ (where $a, b, c \in \mathbb{N}$), where
- A) $a = 21$ B) $a + c = 23$ C) $b + c = 1$ D) $a + b = 22$
8. The value of $\sum_{k=0}^7 \left[\frac{\binom{7}{k}}{\binom{14}{k}} \sum_{r=k}^{14} \binom{r}{k} \binom{14}{r} \right]$, where $\binom{n}{r}$ denotes nC_r , is
- A) 6^7 B) greater than 7^6 C) 8^7 D) greater than 7^8
9. The number of 4 digit natural numbers which contains
- A) exactly 2 distinct digits is 567 B) atleast 2 distinct digits is 8090
C) atleast 2 distinct digits is 577 D) atleast 3 identical digits is 333
10. If $2^{n-1} T_n = {}^{2n}C_0 + 3({}^{2n}C_2) + 3^2({}^{2n}C_4) + \dots + 3^n({}^{2n}C_{2n})$, $n \in \mathbb{N}$ ($n > 2$), then
- A) T_n, T_{n+1}, T_{n+2} are in AP B) $T_n, 2T_{n+1}, T_{n+2}$ are in AP
C) $T_3 = 52$ D) $T_4 = 194$
11. Let k be the number of 10 digit numbers, formed using the digits 0,1,2...9 (with repetition allowed), such that the product of any two consecutive digits in the number is a prime number, then

A) k is a perfect square of a natural number

B) $\frac{k}{2}$ is a perfect square of a natural number

C) $2k$ is a perfect cube of a natural number

D) $4k$ is a perfect cube of a natural number

12. Let p_n denotes the number of permutations of n distinct things taken all at a time and

$x_n = {}^{n+5}C_4 - \left(\frac{143}{96}\right) \left(\frac{p_{n+5}}{p_{n+3}}\right)$ (where $n \in N$). The possible value of n for which x_n is negative, can be

A) 1

B) 2

C) 3

D) 4

SECTION - 3 (Maximum Marks : 24)

This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**

For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer. If the numerical value has more than two decimal places **truncate/round-off** the value to **TWO** decimal places.

Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +4 If ONLY the correct numerical value is entered as answer.

Zero Marks: 0 In all other cases.

13. Let N be the number of positive integral divisors of 1998, then sum of digits of N is ____

14. The number of different seven digit numbers that can be written using only the three digits 4,2,6 with the condition that digit 6 occurs twice in each number is ____

15. Let X be the number of linear permutations of all the letters of the word *SBISMART* in which the two S 's are always together, Y be the number of linear permutations of all the letters of the word *SBISMART* in which the two S 's are never together, then $\frac{Y}{X}$ is ____

16. Let P be ways in which 5 Boys and 5 Girls form a circle such that Boys and Girls are alternate Q be ways in which 5 Boys and 5 Girls form a line such that Boys and Girls are alternate. Then the value of $\frac{Q}{P} - 10$ is ____

17. Let S be the set of all 2×2 matrices formed by using the elements of the set $\{0,1,-1\}$. Then the number of singular matrices which belong to the set S is ____

18. Let P be the number of distinct ways of painting six sides of a cube with 6 distinct colours, where you would regard two cube colourings the same if one can be obtained from the other by rotation, then sum of digits of P is ____

PHYSICS

MAX.MARKS: 66

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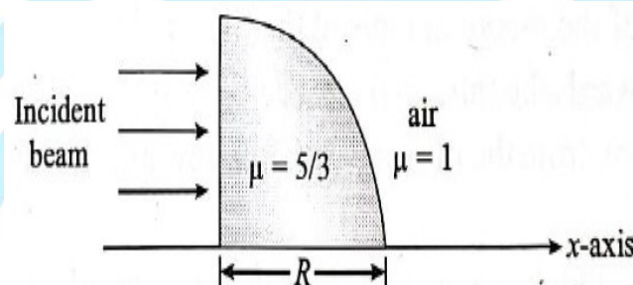
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Full Marks : +3 If only the correct option is chosen.

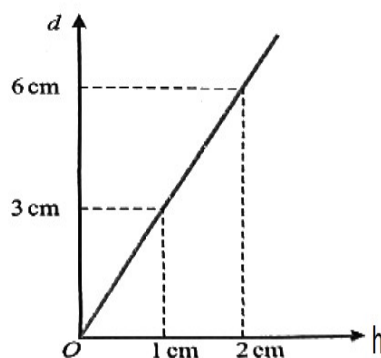
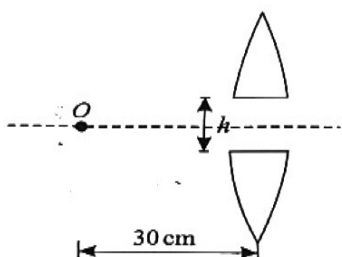
Zero Marks: 0 If none of the option is chosen.(i.e the question is un answered)

Negative Marks: -1 In all other cases.

19. A uniform, horizontal parallel beam of light is incident upon a prism as shown. The prism is in the shape of a quarter cylinder, of radius 5 cm, and has refractive index $5/3$. The width of the region at which the incident rays after normal incidence on plane surface and subsequent refraction at curved surface intersect on x axis is (Neglect the ray which travels along x-axis)

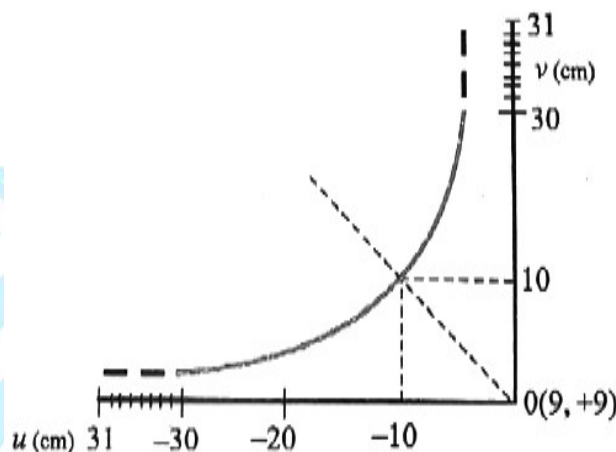


- A) 4 cm B) $\frac{5}{4}$ cm C) $\frac{9}{4}$ cm D) $\frac{25}{4}$ cm
20. A convex lens cut symmetrically into two equal halves and separated laterally by a distance h as shown in figure. A point object O is placed at a distance 30 cm, from the lens halves. The cut parts of the lens forms two real images separated by a distance ' d '. A graph ' d ' versus ' h ' is plotted as shown in figure. The focal length of the lens should be

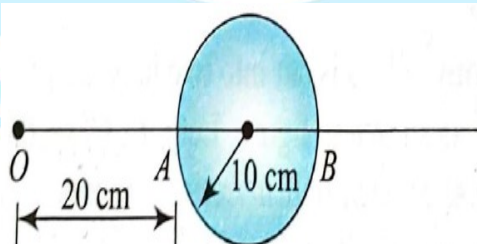


- A) 20 cm B) 50 cm C) 40 cm D) 30 cm

21. A lens forms a diminished and erect image of an object. The magnification is $\frac{1}{4}$. Find ratio of distances between object and focus and focus and image.
- A) 4 : 1 B) 1 : 4 C) 8 : 1 D) 2 : 1
22. The graph shows relationship between object distance and image distance for an equiconvex lens. Then, focal length of the lens is



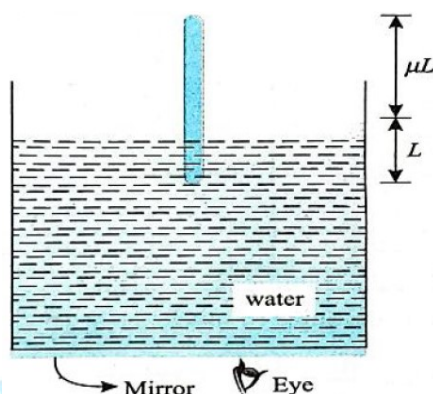
- A) $0.50 \pm 0.05 \text{ cm}$ B) $0.50 \pm 0.10 \text{ cm}$
 C) $5.00 \pm 0.05 \text{ cm}$ D) $5.00 \pm 0.10 \text{ cm}$
23. A glass sphere of radius $R = 10 \text{ cm}$ is kept inside water. A point object O is placed at 20 cm from A as shown in the figure. Find the position of image from point B and nature of the image when seen from other side of the sphere. Given $\mu_g = \frac{3}{2}$ and $\mu_w = \frac{4}{3}$



- A) 200 cm, virtual B) 100 cm, real
 C) 100 cm, virtual D) 300 cm, virtual

24. What is the length of the image of the rod in mirror, according to the observer in air?

(refractive index of the water is μ)



A) $\mu L + L$

B) $L + \frac{L}{\mu}$

C) $L\mu + \frac{L}{\mu}$

D) $L\mu - \frac{L}{\mu}$

SECTION - 2 (Maximum Marks : 24)

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Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options.

Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.

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Negative Marks: -2 In all other cases.

25. A convex lens forms a real image of an object with magnification 0.5. If the object is displaced by 20 cm along the principal axis, a real image equal to the size of the object is formed:

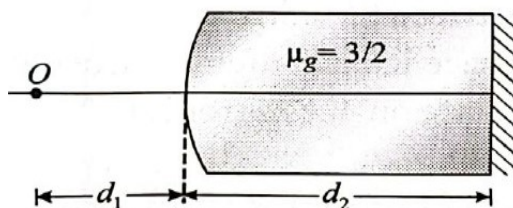
A) the focal length of lens is 20 cm

B) the distance of the image from the lens initially is 25 cm

C) the distance of the object from the lens initially is 60 cm

D) the distance of the image finally from the lens is 30 cm

26. In the figure, a point object O is placed in air on the principal axis. The radius of curvature of the spherical surface is 60 cm. I_f is the final image formed after all the refractions and reflections:



- A) If $d_1 = 120$ cm, then the ' I_f ' is formed on 'O' for any value of d_2
- B) If $d_1 = 240$ cm, then the ' I_f ' is formed on 'O' only if $d_2 = 360$ cm
- C) If $d_1 = 240$ cm, then the ' I_f ' is formed on 'O' for all values of d_2
- D) If $d_1 = 240$ cm, then the ' I_f ' cannot be formed on 'O'
27. A plano-convex lens of focal length 20 cm has its plane side silvered.
- A) The radius of curvature of curved surface of given plano-convex lens is equal to half of radius of curvature of a surface of equiconvex lens of focal length 20 cm
- B) An object placed at 15 cm on the axis on the convex side of silvered plano-convex lens gives rise to an image at a distance of 30 cm from it
- C) An object placed at a distance of 20 cm on the axis on the convex side of silvered plano-convex lens gives rise to an image at 40 cm from it
- D) Silvered plano-convex lens acts as a concave mirror of focal length 10 cm.
28. A plano-convex lens is made of a material of refractive index μ . When a small object is placed 30 cm away from the curved surface of the lens, an image of double the size of the object is produced. Due to reflection from the convex surface of the lens, another faint image is observed at a distance of 10 cm away from the lens. Which of the following statement (s) is (are) true?
- A) The refractive index of the lens is 2.5.
- B) The radius of curvature of the convex surface is 45 cm.
- C) The faint image is erect and real.
- D) The focal length of the lens is 20 cm.

29. A convex lens is placed in the XYZ coordinate system such that its optical centre is at the origin and principal axis is along the x-axis. The focal length of the lens is 20 cm. A point object has been placed at the point $(-40 \text{ cm}, +1 \text{ cm}, -1 \text{ cm})$. Which of the following are correct about coordinates of the image?

- A) $x = 40 \text{ cm}$ B) $y = +1 \text{ cm}$ C) $z = +1 \text{ cm}$ D) $z = -1 \text{ cm}$

30. A ray of light, travelling in a transparent medium fall on a surface separating the medium from air at an angle of incidence of 45° . The ray undergoes total internal reflection. If μ is the refractive index of the medium with respect to air, select the possible value(s) of μ from the following:

- A) 1.3 B) 1.4 C) 1.5 D) 1.6

SECTION - 3 (Maximum Marks : 24)

This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**

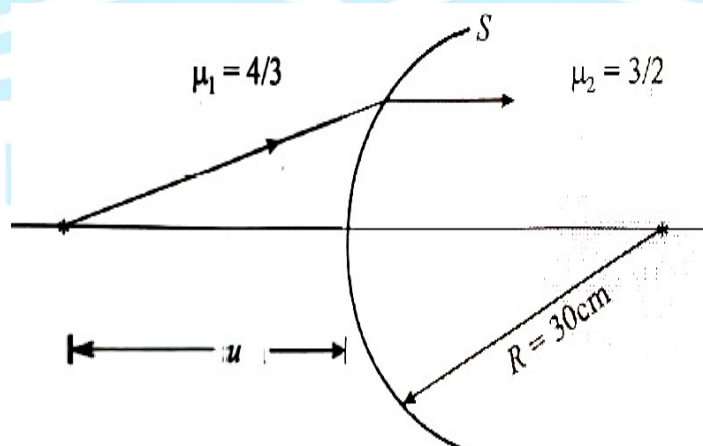
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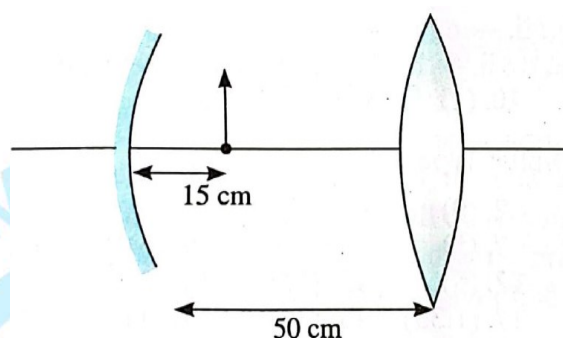
Full Marks: +4 If ONLY the correct numerical value is entered as answer.

Zero Marks: 0 In all other cases.

31. A spherical surface S separates two media 1 and 2 as shown in figure. Find, where an object O is placed in medium – I, so that the light rays from object after refraction becomes parallel to optic axis of this system.



32. Consider a concave mirror and a convex lens (refractive index = 1.5) of focal length 10 cm each, separated by a distance of 50 cm in air (refractive index = 1) as shown in the figure. An object is placed at a distance of 15 cm from the mirror. Its erect image formed by this combination has magnification M_1 . When the set-up is kept in a medium of refractive index $\frac{7}{6}$, the magnification becomes M_2 . The magnitude $\left| \frac{M_2}{M_1} \right|$ is _____.



33. A pin is placed 10 cm in front of a convex lens of focal length 20 cm, made of material having refractive index 1.5. The surface of the lens farther away from the pin is silvered and has a radius of curvature 22 cm. Determine the position of the final image from lens (in cm).
34. The distance between an object and the screen is 100 cm. A lens produces an image on the screen when the lens is placed at either of the positions 40 cm apart. The power of the lens is nearly ___ D.
35. The magnification of an object placed in front of a convex lens is +2. The focal length of the lens is 2.0 metres. Find the distance by which object has to be moved to obtain a magnification of -2 (in metres).
36. A container of uniform cross-section has a height of 14 m. Up to what height (in metre) water of refractive index $\frac{4}{3}$ should be filled inside the container so that container seems to be half filled for normal viewing.

CHEMISTRY**MAX.MARKS: 66****SECTION – 1 (Maximum Marks: 18)**

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Negative Marks: -1 In all other cases.

37. What is true about lanthanides?

- i) They form M_2O_3 & $M(OH)_3$ type compounds under appropriate conditions and their aqueous solutions are basic
- ii) Standard potential of Ce^{4+}/Ce^{3+} is largely positive and thus thermodynamically Ce^{+4} can oxidise water but still in practice it can't oxidise water as the reaction is slow kinetically
- iii) The decrease in ionic radii of M^{+3} ions from La^{+3} to Lu^{+3} is regular but the decrease in atomic radii from La to Lu is not uniform

A) I & ii B) ii & iii C) I & iii D) I, ii & iii

38. Which of the following arrangements does not represent the correct order of the property stated against it?

- A) $Sc < Ti < Cr < Mn$: number of oxidation states
- B) $Ni^{2+} < Co^{2+} < Fe^{2+} < Mn^{2+}$: ionic size
- C) $Co^{3+} < Fe^{3+} < Cr^{3+} < Sc^{3+}$: stability in aqueous solution
- D) $V^{2+} < Cr^{2+} < Mn^{2+} < Fe^{2+}$ paramagnetic behaviour

39. The occurrence of Zirconium & Hafnium together in nature & for the difficulty in their separation is best attributed to?

- A) almost similar melting & boiling points which is a consequence of lanthanoid contraction
- B) almost similar ionization enthalpies & electron gain enthalpies which is a consequence of lanthanoid contraction
- C) almost similar density which is a consequence of lanthanoid contraction
- D) almost similar atomic radii which is a consequence of lanthanoid contraction

40. What is correct about electronic configuration of lanthanides & Actinides?
- i) 5 f electrons can participate in bonding to a far greater extent than 4f electrons
 ii) only 8 of the lanthanides have 7 or more electrons in their 4f level in +3 oxidation state.
- A) only I B) only ii C) both I & ii D) neither I nor ii
41. Which of the property of interstitial compounds has the same behavior as that of the original element/metal?
- A) Malleability B) Ductility
 C) Electrical conductance D) Hardness
42. Anhydrous ferric chloride is prepared by
- A) Dissolving ferric hydroxide in dilute HCl
 B) Dissolving ferric hydroxide in conc. HCl
 C) By passing dry Cl_2 gas over heated scrap iron
 D) By passing dry HCl gas over heated scrap iron

SECTION - 2 (Maximum Marks : 24)

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Negative Marks: -2 In all other cases.

43. Compound 'P' (orange coloured solid) on heating gives compound 'Q' (green colour), 'R' (a dry colourless gas) and water. 'R' on heating with Mg gives a compound 'S' (white solid) which on hydrolysis gives a colourless pungent smelling gas 'T'. Gas 'T' gives white fumes with concentrated HCl. Choose the correct statement(s)?

- A) Q is an amphoteric oxide
- B) aqueous solution of gas T can be titrated with HCl using phenolphthalein as indicator
- C) gas R is a diatomic molecule and its single bond strength is more than C – C bond
- D) gas R is produced during elemental analysis of organic compounds by Kjeldahl's method
44. d block metals and their compounds are used as catalysts. Identify all those combinations which are correctly matched?
- A) TiCl_4 – Adams catalyst in reduction
- B) PdCl_2 – Wacker process for converting C_2H_4 to CH_3CHO
- C) Pt/PtO_2 – Used as the Ziegler-Natta catalyst in polythene production
- D) Cu – In the preparation of $(\text{CH}_3)_2\text{SiCl}_2$
45. Choose the correct statement(s)
- A) The correct order of ionisation energy is: $\text{Au} > \text{Cu} > \text{Ag}$
- B) in 3d series, Cr and Cu have more IE_2 than their neighbors
- C) FeX_2 and FeY_3 are known when $\text{X} = \text{F}, \text{Cl}, \text{Br}, \text{I}$ and $\text{Y} = \text{F}, \text{Cl}, \text{Br}$ respectively
- D) The melting point of Cu, Ag and Au follow the order: $\text{Cu} > \text{Au} > \text{Ag}$
46. Pick out the correct statement(s):
- A) Ionisation enthalpies of lanthanoids are comparable to those of calcium.
- B) finely divided URANIUM when treated with boiling water gives a mixture of oxide and hydride.
- C) Colour of KMnO_4 is because of LMCT, Ligand to Metal Charge Transfer.
- D) KMnO_4 is manufactured by the fusion of pyrolusite ore with KOH in presence of air or KNO_3 , followed by electrolytic oxidation.
47. Acidified potassium permanganate solution is decolourised by
- A) Bleaching powder
- B) White vitriol
- C) Mohr's salt
- D) Glauber's salt

48. In context with the d block elements, which of the following statements is incorrect?
- A) In addition to the normal oxidation states, the zero-oxidation state is also shown by these elements in complexes
- B) In the highest oxidation states, the transition metal shows basic character and form cationic complexes.
- C) in 3d series, Mn has open cubic structure and Zn has poor metallic bonding
- D) stability of maximum oxidation state increases down the group, for example, $V_2O_5 < Nb_2O_5 < Ta_2O_5$

SECTION - 3 (Maximum Marks : 24)

This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**

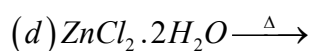
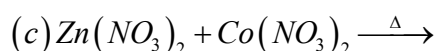
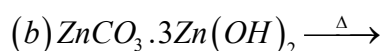
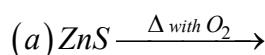
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Full Marks: +4 If ONLY the correct numerical value is entered as answer.

Zero Marks: 0 In all other cases.

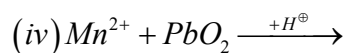
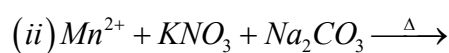
49. How many of the following statements are correct?
- (i) Orange solution of sodium dichromate, can be crystallized as $Na_2Cr_2O_7 \cdot 2H_2O$.
- (ii) Cu^{+2} ion disproportionate in aqueous solution into Cu^{+3} and Cu^{+}
- (iii) Cr^{+2} can liberate hydrogen gas from a dilute acid.
- (iv) CrF_6 exists but MnF_7 does not.
- (v) on strong heating both $KMnO_4$ and $K_2Cr_2O_7$ evolve O_2 gas.
50. Spin only magnetic moment of the complex, $[Cu(NH_3)_4]SO_4$ in the units of Bohr-magnetons is?
- (give answer up to 2 decimals)
51. How many of the following reactions produces white residue (colour is observed after cooling).



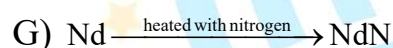
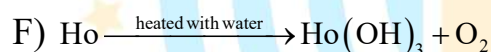
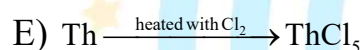
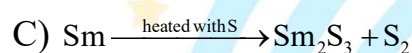
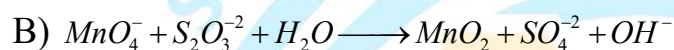
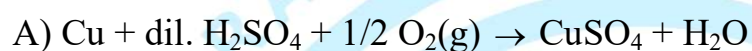
52. Total number of transition metals which have only one oxidation state other than zero.

Fe, Cd, Zn, Hg, Ag, Au, Cu, Ni

53. Find out total number of reaction(s) under given condition(s), which convert Mn^{2+} into MnO_4^-



54. Select the CORRECT chemical changes?



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